Radio Astronomy
A Derivation Handbook

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Personal Reference Notes

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Chapter 1

Essential mathematics

1.1 Vector Calculus Rules

1.1.1 Triple Products

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \tag{1.1}$$

Note that the ABC cyclic order is maintained, this is the scaler tripler product.

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \tag{1.2}$$

This is the "BAC-CAB" rule, the vector triple product (1.2).

Chapter 2

The Radio Astronomer's Electromagnetism

2.1 Maxwell's Equations

Maxwell's equations in SI units are

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \qquad \text{(Gauss's law for electricity)} \tag{2.1a}$$

$$\nabla \cdot \mathbf{B} = 0, \qquad \text{(Gauss's law for magnetism)} \tag{2.1b}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \qquad \text{(Faraday's law of induction)} \tag{2.1c}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \qquad \text{(Ampère–Maxwell law)} \tag{2.1d}$$

where $\mathbf{E}$ is the electric field, $\mathbf{B}$ is the magnetic field, ρ is the electric charge density, $\mathbf{J}$ is the electric current density, ϵ_0 is the vacuum permittivity, and μ_0 is the vacuum permeability.

Another useful equation to remember is the **continuity equation**, which expresses the local conservation of electric charge:

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0} \tag{2.2}$$

where ρ is the electric charge density and $\mathbf{J}$ is the electric current density, defined as

$$\mathbf{J} = \rho \mathbf{v}, \tag{2.3}$$

where $\mathbf{v}$ is the velocity of the moving charge distribution. The SI units of $\mathbf{J}$ are $\mathrm{A\,m^{-2}}$.

2.2 Deriving the Electromagnetic Wave Equation

2.2.1 The Wave Equation

The **wave equation** is a second-order partial differential equation (PDE) that describes the propagation of disturbances through space and time. Its general three-dimensional form is

$$\boxed{\nabla^2 \mathbf{R} = \frac{1}{v^2} \frac{\partial^2 \mathbf{R}}{\partial t^2}} \tag{2.4}$$

where $\mathbf{R}(\mathbf{r}, t)$ represents a vector displacement field, $\mathbf{r}$ is the position vector, v is the wave propagation speed, and ∇^2 is the Laplacian operator.

In one spatial dimension, the wave equation becomes

$$\boxed{\frac{\partial^2 x}{\partial s^2} = \frac{1}{v^2}\frac{\partial^2 x}{\partial t^2}} \tag{2.5}$$

where $x(s, t)$ represents the displacement at position s and time t.

It relates the second space derivative of a function, to the second time derivative. The v is the wave speed. For deriving the EM wave equation, we just need to get an equation in the shape of Equation 2.6 from Maxwell's equations.

2.2.2 Deriving The EM Wave Equation

To derive the EM wave equation in free space, we look at Maxwell's Equations 2.1, and the Classical Wave Equation 2.6.

Our goal is:

$$\boxed{\nabla^2 \mathbf{E} = \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2}} \tag{2.6}$$

or equally in terms $\mathbf{B}$.

Lets attack the LHS, we can take the curl of $\nabla \times \mathbf{E}$ (Faraday's law, Equation 2.1c).Writing this we have

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla \times (-\frac{\partial \mathbf{B}}{\partial t}) \tag{2.7}$$

This is a daunting vector triple product we recognize as the "BAC-CAB", the identify is $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$ (See Vector Calculus rule (1.2).

In our case this becomes:

$$\nabla(\nabla \cdot \mathbf{E}) - E(\nabla \cdot \nabla) = \nabla \times (-\frac{\partial \mathbf{B}}{\partial t}) \tag{2.8}$$

We recognize $\nabla \cdot \mathbf{E}$ as Gauss's Law (Eq 2.1a, it must be $= 0$, as $\rho = 0$ in free space. We further reconize the Laplacian operator $\nabla \cdot \nabla = \nabla^2$, we also have move it to the correct side of $\mathbf{E}$, we also move the partial time derivative outside of the curl.

$$-\nabla^2 \mathbf{E} = -\frac{\partial}{\partial t} \tag{2.9}$$

The curl of $\mathbf{B}$ is one of Maxwell's laws! It is $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$ (Ampere-Maxwell Law Eq 2.1d). Since we are in free-space $\mathbf{J} = 0$, and we have

$$\boxed{\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}} \tag{2.10}$$

and the speed of light is given by:

$$\boxed{c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}} \tag{2.11}$$

2.3 EM Wave Fundamentals

2.3.1 Plane Electromagnetic Waves

A simple solution to the electromagnetic wave equation is a plane wave propagating in the $+x$ direction,

$$\boxed{\mathbf{E}(x,t) = \mathrm{Re}\left[\mathbf{E}_0 e^{i(kx-\omega t)}\right]} \tag{2.12}$$

where $\mathbf{E}_0$ is the complex electric field amplitude, k is the angular wavenumber, and ω is the angular frequency.

The angular wavenumber is defined as

$$\boxed{k = \frac{2\pi}{\lambda}} \tag{2.13}$$

where λ is the wavelength. Physically, k describes how rapidly the phase changes with position, measured in radians per metre.

Similarly, the angular frequency is

$$\boxed{\omega = 2\pi f = \frac{2\pi}{T}} \tag{2.14}$$

where f is the frequency and T is the period. Physically, ω describes how rapidly the phase changes with time, measured in radians per second.

In three dimensions, the plane wave becomes

$$\mathbf{E}(\mathbf{r},t) = \mathrm{Re}\left[\mathbf{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}\right], \tag{2.15}$$

where $\mathbf{k}$ is the wavevector, pointing in the direction of propagation, with magnitude $|\mathbf{k}| = k$.

2.3.2 Relationship Between Electric and Magnetic Fields

Transverse Nature of EM Waves

We can show that electromagnetic waves are transverse using Gauss's laws in free space,

$$\boxed{\nabla\cdot\mathbf{E} = 0, \qquad \nabla\cdot\mathbf{B} = 0.} \tag{2.16}$$

Consider a plane wave of the form

$$\mathbf{E}(\mathbf{r},t) = \mathrm{Re}\left[\mathbf{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}\right], \tag{2.17}$$

where $\mathbf{k}$ is the wavevector and $\mathbf{E}_0$ is the constant electric field amplitude.

Taking the divergence of the complex plane-wave representation gives

$$\nabla\cdot\left(\mathbf{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}\right) = i(\mathbf{k}\cdot\mathbf{E}_0)e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}. \tag{2.18}$$

From Gauss's law, this must equal zero in free space. Since the exponential is never zero, we obtain

$$\boxed{\mathbf{k}\cdot\mathbf{E}_0 = 0}. \tag{2.19}$$

Repeating the same calculation for the magnetic field, using $\nabla \cdot \mathbf{B} = 0$, gives

$$\boxed{\mathbf{k} \cdot \mathbf{B}_0 = 0}. \tag{2.20}$$

Therefore,

$$\boxed{\mathbf{k} \cdot \mathbf{E}_0 = 0, \qquad \mathbf{k} \cdot \mathbf{B}_0 = 0} \tag{2.21}$$

Both the electric and magnetic fields are perpendicular to the direction of wave propagation, $\mathbf{k}$. This is why electromagnetic plane waves in free space are called **transverse waves**.

and Faraday's law gives

$$\boxed{\mathbf{B}_0 = \frac{1}{\omega}\mathbf{k} \times \mathbf{E}_0} \tag{2.22}$$

Thus, the electric field, magnetic field, and propagation direction are mutually perpendicular.

In free space, their amplitudes satisfy

$$\boxed{E_0 = cB_0} \tag{2.23}$$

where c is the speed of light.

2.3.3 Phase and Group Velocity

The phase of a plane electromagnetic wave is the argument of the complex exponetial.

$$\boxed{\phi = kx - \omega t}. \tag{2.24}$$

To follow a point of constant phase, we set $d\phi = 0$, giving

$$k\,dx - \omega\,dt = 0. \tag{2.25}$$

Therefore, the phase velocity is

$$\boxed{v_{\text{ph}} = \frac{\omega}{k}} \tag{2.26}$$

The phase velocity describes the speed at which a point of constant phase, such as a wave crest, propagates. A wave packet can be constructed by superposing waves with slightly different frequencies and wavenumbers. The speed of the resulting envelope is called the **group velocity**,

$$\boxed{v_{\text{g}} = \frac{d\omega}{dk}} \tag{2.27}$$

The group velocity describes the propagation of the wave packet envelope. In a transparent, weakly absorbing medium it is often associated with the transport of energy.

In free space, the dispersion relation is

$$\boxed{\omega = ck} \tag{2.28}$$

and consequently,

$$\boxed{v_{\text{ph}} = v_{\text{g}} = c} \tag{2.29}$$

2.3.4 Dispersion

Dispersion occurs when the relationship between angular frequency ω and wavenumber k is not linear.

In a dispersive medium, waves of different frequencies can propagate with different phase and group velocities.

Dispersion in Free Space

In classical vacuum electrodynamics, the dispersion relation is

$$\omega^2 = c^2 k^2. \tag{2.30}$$

Because $\omega = ck$ for positive-frequency waves, both phase and group velocities are independent of frequency.

Therefore, free space is a non-dispersive medium.

Electromagnetic Waves in Matter

Maxwell's equations remain valid in matter, but the electromagnetic response of the material must be included.

For macroscopic electromagnetic fields, we introduce the electric displacement field $\mathbf{D}$ and magnetic field intensity $\mathbf{H}$,

$$\mathbf{D} = \epsilon \mathbf{E}, \qquad \mathbf{B} = \mu \mathbf{H}, \tag{2.31}$$

where ϵ and μ are the permittivity and permeability of a linear, isotropic medium.

In a homogeneous, source-free, lossless, non-dispersive medium, Maxwell's equations give the wave equation

$$\boxed{\nabla^2 \mathbf{E} = \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}} \tag{2.32}$$

The wave propagation speed is therefore

$$\boxed{v = \frac{1}{\sqrt{\mu\epsilon}}} \tag{2.33}$$

The refractive index is defined as

$$\boxed{n = \frac{c}{v}} \tag{2.34}$$

For a frequency-dependent material response, the refractive index can depend on frequency,

$$n = n(\omega). \tag{2.35}$$

For a transparent, isotropic medium, the dispersion relation is then

$$\boxed{k(\omega) = \frac{n(\omega)\omega}{c}} \tag{2.36}$$

The phase velocity becomes

$$\boxed{v_{\rm ph} = \frac{c}{n(\omega)}} \tag{2.37}$$

while the group velocity is

$$\boxed{v_{\rm g} = \frac{c}{n(\omega) + \omega \frac{{\rm d}n}{{\rm d}\omega}}} \tag{2.38}$$

Thus, when the refractive index depends on frequency, the phase and group velocities are generally different.

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2.4 Dispersion

Dispersion is the phenomenon in which waves of different frequencies propagate at different speeds. More precisely, the relation between angular frequency ω and wavenumber k is called the *dispersion relation*, $\omega = \omega(k)$. A wave packet propagates at the group velocity $v_g = \frac{{\rm d}\omega}{{\rm d}k}$, while individual phase fronts move at $v_{\rm ph} = \omega/k$.

2.4.1 Dispersion in free space

In vacuum, where there are no charges or currents, Maxwell's equations give

$$\nabla \cdot \mathbf{E} = 0, \qquad \nabla \cdot \mathbf{B} = 0, \tag{2.39}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \qquad \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \tag{2.40}$$

Take the curl of Faraday's law:

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}. \tag{2.41}$$

Using $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$, and substituting Ampère–Maxwell's law, we obtain

$$\boxed{\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}.} \tag{2.42}$$

Try a plane wave $\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)}$. Then $\nabla^2 \mathbf{E} = -k^2 \mathbf{E}$ and $\partial_t^2 \mathbf{E} = -\omega^2 \mathbf{E}$, so

$$k^2 = \mu_0 \epsilon_0 \omega^2, \qquad \boxed{\omega = ck}, \qquad c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}. \tag{2.43}$$

Therefore

$$\boxed{v_{\rm ph} = \frac{\omega}{k} = c, \qquad v_g = \frac{{\rm d}\omega}{{\rm d}k} = c.} \tag{2.44}$$

All frequencies propagate at the same speed: vacuum is nondispersive.

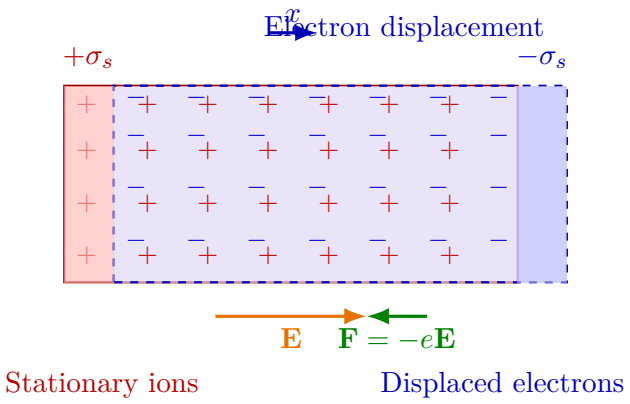


Figure 2.1: A displacement of the electron fluid relative to the stationary ions produces oppositely charged surface layers. The resulting electric field points from the positive sheet toward the negative sheet, exerting a restoring force on the electrons. The displacement is exaggerated for clarity.

2.4.2 The plasma frequency: physical intuition

Imagine a uniform slab of plasma, with electron number density n_e and stationary positive ions. Displace the electron fluid a small distance x to the right relative to the ions. The slab remains neutral in its interior, but thin layers of opposite surface charge appear at its edges, with magnitudes $\sigma_s = n_e e x$ (charge per unit area).

By Gauss's law, $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$. Equivalently, a Gaussian pillbox across a charged sheet gives $\Delta E_\perp = \sigma_s/\epsilon_0$. Combining the fields of the two oppositely charged sheets gives an electric field inside the slab of magnitude

$$E = \frac{\sigma_s}{\epsilon_0} = \frac{n_e e x}{\epsilon_0}. \tag{2.45}$$

For a positive displacement x, the field points to the right, so the force $-eE$ on an electron points to the left: it is a restoring force. Newton's law becomes

$$m_e \frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = -eE = -\frac{n_e e^2}{\epsilon_0} x. \tag{2.46}$$

Thus

$$\boxed{\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \omega_p^2 x = 0, \qquad \omega_p^2 = \frac{n_e e^2}{\epsilon_0 m_e}.} \tag{2.47}$$

The plasma frequency is the natural oscillation frequency of this collective restoring motion.

Electron oscillation versus thermal motion. The electron fluid oscillates harmonically:

$$x(t) = x_0 \cos(\omega_p t). \tag{2.48}$$

Its maximum coherent oscillation velocity is therefore

$$\boxed{v_{\mathrm{osc}} = \omega_p x_0.} \tag{2.49}$$

By the equipartition theorem, each quadratic degree of freedom contributes $\frac{1}{2} k_B T_e$ to the average kinetic energy. For one-dimensional thermal motion,

$$\boxed{\frac{1}{2} m_e \langle v_x^2 \rangle = \frac{1}{2} k_B T_e \quad \Longrightarrow \quad v_{\mathrm{th}} = \sqrt{\frac{k_B T_e}{m_e}}.} \tag{2.50}$$

For example, at $T_e = 10^4\,\mathrm{K}$,

$$v_{\mathrm{th}} \simeq 3.9 \times 10^5\,\mathrm{m\,s^{-1}} \simeq 390\,\mathrm{km\,s^{-1}}. \tag{2.51}$$

The relative importance of the two velocities is

$$\boxed{\frac{v_{\mathrm{osc}}}{v_{\mathrm{th}}} = \frac{\omega_p x_0}{\sqrt{k_B T_e/m_e}}.} \tag{2.52}$$

Unlike the thermal velocity, the oscillation velocity depends on the amplitude of the disturbance. For sufficiently small perturbations, v_{osc} can be much smaller than v_{th}.

Importantly, the **cold-plasma approximation does not require** $v_{\mathrm{osc}} > v_{\mathrm{th}}$. Rather, it assumes that thermal pressure and velocity-spread effects are negligible in the wave dynamics.

For longitudinal waves, this typically requires $kv_{\mathrm{th}} \ll \omega$, so that electrons do not move appreciably across a wavelength during one oscillation. For transverse electromagnetic waves, thermal corrections to the usual cold, unmagnetized plasma dispersion relation are likewise negligible in the appropriate long-wavelength, nonrelativistic regime.

Thus, a plasma may have substantial electron thermal motion and still be accurately described by the cold-plasma approximation.

2.4.3 Dispersion in cold plasma

Now consider a *collisionless, unmagnetized, cold plasma*: a neutral equilibrium with electron density n_e, stationary ions, and negligible electron thermal pressure. "Cold" means we neglect thermal motions in the linearized electron momentum equation; it does not mean the electrons are at absolute zero. Let $e > 0$ be the magnitude of the electron charge, and m_e its mass.

Step 1: Electron response to an electric field. We look for a single-frequency wave, so we write $\mathbf{E} \propto e^{-i\omega t}$. Why? In a linear, time-independent medium, sinusoidal Fourier components evolve independently; complex exponentials make time differentiation especially simple: $\partial_t \to -i\omega$. We take the real part at the end. This is an ansatz for a monochromatic component, not an assumption that every physical pulse has only one frequency.

For small-amplitude waves, the electron velocity $\mathbf{v}$ obeys

$$m_e \frac{\partial \mathbf{v}}{\partial t} = -e\mathbf{E}. \tag{2.53}$$

The magnetic Lorentz force $-e\mathbf{v} \times \mathbf{B}$ is second order in wave amplitude because both $\mathbf{v}$ and the wave magnetic field are first-order perturbations. For harmonic fields proportional to $e^{-i\omega t}$,

$$-i\omega m_e \mathbf{v} = -e\mathbf{E} \quad \Longrightarrow \quad \boxed{\mathbf{v} = -i\frac{e}{m_e\omega}\mathbf{E}.} \tag{2.54}$$

Recall that $\mathbf{J}$ is the **electric current per unit area**, with SI units $\mathrm{A\,m^{-2}}$. For electrons with charge $-e$, number density n_e, and velocity $\mathbf{v}$, the current density is $\boxed{\mathbf{J} = -n_e e\mathbf{v}}$. Hence

$$\boxed{\mathbf{J} = i\frac{n_e e^2}{m_e\omega}\mathbf{E}.} \tag{2.55}$$

Unlike an ordinary Ohmic conductor, the collisionless plasma has a frequency-dependent, imaginary effective conductivity.

Step 2: Derive the wave equation. Use Maxwell's equations with vacuum ϵ_0, μ_0 and the plasma current:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{2.56}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \tag{2.57}$$

For a transverse plane wave ($\mathbf{k} \cdot \mathbf{E} = 0$), curl Faraday's law and substitute Ampère's law:

$$\nabla^2 \mathbf{E} = \mu_0 \frac{\partial \mathbf{J}}{\partial t} + \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \tag{2.58}$$

Using the harmonic current above,

$$\frac{\partial \mathbf{J}}{\partial t} = (-i\omega)\left(i \frac{n_e e^2}{m_e \omega} \mathbf{E}\right) = \frac{n_e e^2}{m_e} \mathbf{E}. \tag{2.59}$$

Therefore

$$\boxed{\nabla^2 \mathbf{E} = \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} + \mu_0 \frac{n_e e^2}{m_e} \mathbf{E}.} \tag{2.60}$$

Substituting $\mathbf{E} = \mathbf{E}_0 e^{i(kz - \omega t)}$ gives

$$-k^2 = -\frac{\omega^2}{c^2} + \mu_0 \frac{n_e e^2}{m_e}. \tag{2.61}$$

Multiply by $c^2 = 1/(\mu_0 \epsilon_0)$, and recognize $\omega_p^2 = n_e e^2/(\epsilon_0 m_e)$ from the preceding section:

$$\boxed{\omega^2 = c^2 k^2 + \omega_p^2}, \qquad \boxed{\omega_p \equiv \sqrt{\frac{n_e e^2}{\epsilon_0 m_e}}}. \tag{2.62}$$

Here ω_p is the electron plasma frequency derived above. Its ordinary-frequency form is $\nu_p = \omega_p/(2\pi)$.

Step 3: Index of refraction, phase and group velocities. The **index of refraction** n is defined as the ratio of the speed of light in vacuum to the phase velocity of a wave in a medium:

$$\boxed{n \equiv \frac{c}{v_{\text{ph}}} = \frac{ck}{\omega}.} \tag{2.63}$$

Using the plasma dispersion relation $\omega^2 = c^2 k^2 + \omega_p^2$, we obtain

$$\boxed{n = \sqrt{1 - \frac{\omega_p^2}{\omega^2}}.} \tag{2.64}$$

Unlike most ordinary materials, a cold plasma has $n < 1$ for propagating electromagnetic waves.

From $k = \sqrt{\omega^2 - \omega_p^2}/c$, the phase velocity is

$$\boxed{v_{\text{ph}} = \frac{\omega}{k} = \frac{c}{n} = \frac{c}{\sqrt{1 - \omega_p^2/\omega^2}}.} \tag{2.65}$$

Differentiate $\omega = \sqrt{c^2k^2 + \omega_p^2}$ to obtain the group velocity:

$$\boxed{v_g = \frac{\mathrm{d}\omega}{\mathrm{d}k} = \frac{c^2 k}{\omega} = cn = c\sqrt{1 - \frac{\omega_p^2}{\omega^2}}.} \tag{2.66}$$

Thus $v_{\mathrm{ph}} v_g = c^2$. Phase velocity can exceed c, but the group velocity of propagating waves is less than c. For $\omega < \omega_p$, k is imaginary and transverse electromagnetic waves are evanescent rather than freely propagating, leading to reflection at a plasma boundary.

Plasma reflection. Recall the cold-plasma dispersion relation,

$$\omega^2 = c^2k^2 + \omega_p^2. \tag{2.67}$$

When $\omega < \omega_p$, the wavenumber k becomes imaginary. Electromagnetic waves can no longer propagate freely through the plasma; instead, they become evanescent and are reflected at the plasma boundary (in the ideal lossless case).

This is why the Earth's ionosphere can reflect radio waves back toward the ground, particularly at frequencies below roughly 10 MHz

2.4.4 Dispersion measure

A radio pulse contains many frequencies. In a dilute plasma, lower frequencies travel more slowly because v_g depends on ω. Suppose $\omega \gg \omega_p$ everywhere along the path. Expand $\sqrt{1-x} \simeq 1 - x/2$ for $x \ll 1$:

$$v_g = c\sqrt{1 - \frac{\omega_p^2}{\omega^2}} \simeq c\left(1 - \frac{\omega_p^2}{2\omega^2}\right). \tag{2.68}$$

Important: We need the *reciprocal* of the group velocity to calculate travel time, not the group velocity itself. Using $1/(1-a) \simeq 1 + a$ for $|a| \ll 1$, the sign changes:

$$\frac{1}{v_g} = \frac{1}{c}\left(1 - \frac{\omega_p^2}{\omega^2}\right)^{-1/2} \simeq \frac{1}{c}\left(1 + \frac{\omega_p^2}{2\omega^2}\right). \tag{2.69}$$

For an element of path length $d\ell$, the extra travel time relative to vacuum is

$$dt_{\mathrm{extra}} = \left(\frac{1}{v_g} - \frac{1}{c}\right) d\ell \simeq \frac{\omega_p^2}{2c\omega^2}\, d\ell. \tag{2.70}$$

Substitute $\omega_p^2 = n_e e^2/(\epsilon_0 m_e)$ and $\omega = 2\pi\nu$:

$$\boxed{\Delta t(\nu) = \frac{e^2}{8\pi^2 \epsilon_0 m_e c \nu^2} \int n_e(\ell)\, d\ell.} \tag{2.71}$$

Define the **dispersion measure** (DM) as the electron column density along the line of sight:

$$\boxed{\mathrm{DM} \equiv \int_0^D n_e(\ell)\, d\ell.} \tag{2.72}$$

Then

$$\boxed{\Delta t(\nu) = \frac{e^2}{8\pi^2 \epsilon_0 m_e c} \frac{\mathrm{DM}}{\nu^2}.} \tag{2.73}$$

For two observing frequencies $\nu_{\rm low} < \nu_{\rm high}$,

$$\boxed{t(\nu_{\rm low}) - t(\nu_{\rm high}) = K_{\rm DM}\,{\rm DM}\left(\nu_{\rm low}^{-2} - \nu_{\rm high}^{-2}\right).} \tag{2.74}$$

A convenient radio-astronomy convention is

$$\boxed{\Delta t\,[{\rm ms}] \simeq 4.1488\,{\rm DM}\,[{\rm pc\,cm^{-3}}]\left[\left(\frac{\nu_{\rm low}}{\rm GHz}\right)^{-2} - \left(\frac{\nu_{\rm high}}{\rm GHz}\right)^{-2}\right].} \tag{2.75}$$

This is the familiar ν^{-2} dispersive sweep seen in pulsars and fast radio bursts. The derivation assumes a cold, weakly dispersive, unmagnetized plasma and neglects cosmological redshift; for cosmological sources, the observed contribution of a plasma element at redshift z is weighted by $1/(1+z)$ when written in terms of its proper electron column.

For an element of path length $d\ell$, the extra travel time relative to vacuum is

$$dt_{\rm extra} = \left(\frac{1}{v_g} - \frac{1}{c}\right) d\ell \simeq \frac{\omega_p^2}{2c\omega^2}\, d\ell. \tag{2.76}$$

Substitute $\omega_p^2 = n_e e^2/(\epsilon_0 m_e)$ and $\omega = 2\pi\nu$:

$$\boxed{\Delta t(\nu) = \frac{e^2}{8\pi^2 \epsilon_0 m_e c \nu^2} \int n_e(\ell)\, d\ell.} \tag{2.77}$$

Define the **dispersion measure** (DM) as the electron column density along the line of sight:

$$\boxed{{\rm DM} \equiv \int_0^D n_e(\ell)\, d\ell.} \tag{2.78}$$

Then

$$\boxed{\Delta t(\nu) = \frac{e^2}{8\pi^2 \epsilon_0 m_e c} \frac{\rm DM}{\nu^2}.} \tag{2.79}$$

For two observing frequencies $\nu_{\rm low} < \nu_{\rm high}$,

$$\boxed{t(\nu_{\rm low}) - t(\nu_{\rm high}) = K_{\rm DM}\,{\rm DM}\left(\nu_{\rm low}^{-2} - \nu_{\rm high}^{-2}\right).} \tag{2.80}$$

A convenient radio-astronomy convention is

$$\boxed{\Delta t\,[{\rm ms}] \simeq 4.1488\,{\rm DM}\,[{\rm pc\,cm^{-3}}]\left[\left(\frac{\nu_{\rm low}}{\rm GHz}\right)^{-2} - \left(\frac{\nu_{\rm high}}{\rm GHz}\right)^{-2}\right].} \tag{2.81}$$

This is the familiar ν^{-2} dispersive sweep seen in pulsars and fast radio bursts.

Cosmological redshift correction. For a plasma element at redshift z, the local wave frequency is $\nu_{\rm local} = (1+z)\nu_{\rm obs}$, while the observed time interval is stretched by a factor $(1+z)$. Together, these give a net factor of $1/(1+z)$.

Thus, for a cosmological source,

$$\boxed{\Delta t_{\rm obs}(\nu_{\rm obs}) = \frac{K_{\rm DM}}{\nu_{\rm obs}^2} \int_0^{z_s} \frac{n_e(z)}{1+z}\, d\ell_{\rm proper}.} \tag{2.82}$$

We can therefore define an observed dispersion measure,

$$\boxed{\mathrm{DM}_{\rm obs} = \int_0^{z_s} \frac{n_e(z)}{1+z}\, d\ell_{\rm proper}.} \tag{2.83}$$

Chapter 3

Radio Scattering

3.1 Scattering Basics, and the $\alpha - 4$ Scattering Index

Scattering causes radiation to follow different paths through an inhomogeneous plasma. Since these paths have different lengths, radiation emitted simultaneously can arrive at different times, broadening the observed pulse.

We will derive the frequency dependence of this broadening, assuming a thin scattering screen and small scattering angles.

How a Plasma Bends Light

Recall that the refractive index of a cold plasma is:

$$n = \sqrt{1 - \frac{\omega_p^2}{\omega^2}} \tag{3.1}$$

and that $w_p =$, when w we can write:

$$n \simeq 1 - \frac{n_e e^2}{2\epsilon_0 m_e \omega^2}. \tag{3.2}$$

Therefore,

$$\boxed{n - 1 \propto n_e \nu^{-2}.} \tag{3.3}$$

If the electron density changes across the wavefront, different parts of the wave travel at different phase velocities. This causes the wavefront to rotate, changing the direction of propagation.

We can derive this using Snell's law.

$$n_1 \sin\phi_1 = n_2 \sin(\phi_2), \tag{3.4}$$

where ϕ is measured relative to the normal of the boundary.

Consider two adjacent regions with refractive indices n and $n+dn$. A ray crossing their boundary obeys

$$n \sin\phi = (n + dn)\sin(\phi + d\phi), \tag{3.5}$$

Expanding to first order,

$$n \sin\phi \simeq n \sin\phi + dn \sin\phi + n \cos\phi \, d\phi. \tag{3.6}$$

Rearranging,

$$d\phi = -\frac{\tan\phi}{n} \, dn. \tag{3.7}$$

Now suppose the refractive index varies in the transverse x direction, while the ray travels approximately along z. For a nearly horizontal refractive-index boundary, the angle ϕ is measured relative to the x direction.

In the small-angle approximation, the ray's angle relative to z is $\theta = \pi/2 - \phi$, so $d\theta = -d\phi$. Furthermore,

$$\tan\phi \simeq \frac{dz}{dx}, \qquad dn = \frac{\partial n}{\partial x} \, dx. \tag{3.8}$$

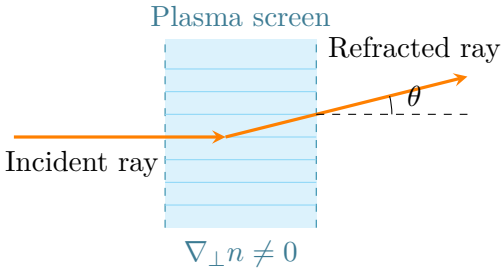
Substituting gives

$$d\theta \simeq \frac{1}{n} \frac{\partial n}{\partial x} \, dz. \tag{3.9}$$

Since $n \simeq 1$ in a tenuous plasma,

$$\boxed{\frac{d\theta}{dz} \simeq \frac{\partial n}{\partial x}.} \tag{3.10}$$

This is the small-angle ray-bending equation: a transverse gradient in refractive index changes the direction of propagation.



For a thin screen, integrate the ray-bending equation through the plasma:

$$\theta \simeq \int \frac{\partial n}{\partial x} \, dz. \tag{3.11}$$

Substituting the plasma refractive index,

$$\theta \simeq -\frac{e^2}{2\epsilon_0 m_e \omega^2} \int \frac{\partial n_e}{\partial x} \, dz. \tag{3.12}$$

Define the electron column density

$$N_e(x) = \int n_e(x, z) \, dz. \tag{3.13}$$

Then

$$\boxed{\theta \simeq -\frac{e^2}{2\epsilon_0 m_e \omega^2}\frac{\partial N_e}{\partial x}.} \tag{3.14}$$

For a fixed transverse electron-density structure,

$$\boxed{\theta \propto \nu^{-2}.} \tag{3.15}$$

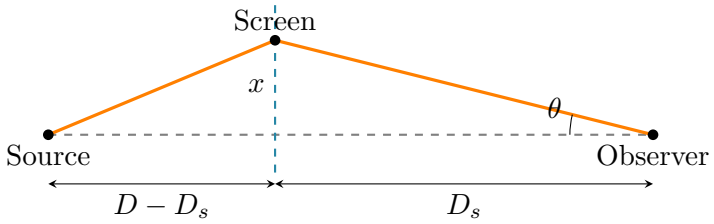
Lower-frequency radio waves are therefore deflected more strongly by the same plasma structure.

From Scattering Angle to Time Delay

We now need to determine how much longer a scattered ray travels compared with an unscattered ray.

Consider a source and observer separated by distance D. A thin scattering screen is located a distance D_s from the observer.

A ray passes through the screen at transverse displacement x from the direct line of sight.



Using the Pythagorean theorem, the length of the scattered path is

$$L_{\text{scatt}} = \sqrt{(D - D_s)^2 + x^2} + \sqrt{D_s^2 + x^2}. \tag{3.16}$$

The direct path has length

$$L_{\text{direct}} = D. \tag{3.17}$$

For small scattering angles, $x \ll D_s$ and $x \ll D - D_s$. We can use the Taylor expansion

$$\sqrt{a^2 + x^2} = a\sqrt{1 + \frac{x^2}{a^2}} \simeq a + \frac{x^2}{2a}. \tag{3.18}$$

Therefore,

$$L_{\text{scatt}} \simeq (D - D_s) + \frac{x^2}{2(D - D_s)} + D_s + \frac{x^2}{2D_s} \tag{3.19}$$

$$= D + \frac{x^2}{2}\left(\frac{1}{D - D_s} + \frac{1}{D_s}\right). \tag{3.20}$$

The excess path length is thus

$$\boxed{\Delta L = \frac{x^2 D}{2D_s(D - D_s)}.} \tag{3.21}$$

From the geometry, the observed angle between the scattered ray and the direct line of sight is

$$\theta \simeq \frac{x}{D_s}. \tag{3.22}$$

Substituting $x \simeq D_s\theta$,

$$\boxed{\Delta L \simeq \frac{D_s D}{2(D - D_s)}\theta^2.} \tag{3.23}$$

The additional geometric travel time is $\tau_{\rm scatt} = \Delta L/c$, giving

$$\boxed{\tau_{\rm scatt} \simeq \frac{D_s D}{2c(D - D_s)}\theta^2.} \tag{3.24}$$

Thus, for a fixed scattering geometry,

$$\boxed{\tau_{\rm scatt} \propto \theta^2.} \tag{3.25}$$

The Scattering Index

We have established two relationships:

$$\theta \propto \nu^{-2}, \qquad \tau_{\rm scatt} \propto \theta^2. \tag{3.26}$$

Combining them,

$$\tau_{\rm scatt} \propto (\nu^{-2})^2. \tag{3.27}$$

Therefore,

$$\boxed{\tau_{\rm scatt} \propto \nu^{-4}.} \tag{3.28}$$

Defining the scattering index α by

$$\tau_{\rm scatt} \propto \nu^{\alpha}, \tag{3.29}$$

we obtain

$$\boxed{\alpha = -4.} \tag{3.30}$$

This result assumes that the characteristic transverse plasma structure responsible for deflection is independent of frequency.

In a turbulent plasma, different frequencies can probe different spatial scales. For an ideal Kolmogorov electron-density fluctuation spectrum, the strong-scattering prediction is instead

$$\boxed{\alpha = -\frac{22}{5} = -4.4.} \tag{3.31}$$

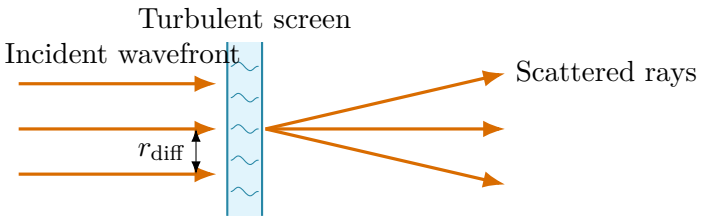
3.2 the Kolmogorov Scattering Index $\alpha = -4.4$

Radio waves propagating through a turbulent ionized medium encounter electron-density fluctuations. These fluctuations impose a spatially varying phase on the wavefront. The wavefront develops structure on a characteristic transverse scale, called the *diffractive scale* $r_{\rm diff}$. Diffraction by this structure produces an angular spread $\theta_{\rm scatt} \sim \lambda/(2\pi r_{\rm diff})$. Different paths then arrive with different geometric delays, producing pulse broadening.

The goal is to derive the frequency dependence

$$\boxed{\tau_{\rm scatt} \propto \nu^{-22/5} \equiv \nu^{-4.4}} \tag{3.32}$$

for a Kolmogorov electron-density spectrum, under the standard assumptions of a thin screen, strong scattering, and an inertial range extending across the diffractive scale.



3.3 Phase delay produced by a plasma

For a cold, unmagnetized plasma with $\omega \gg \omega_p$,

$$n \simeq 1 - \frac{\omega_p^2}{2\omega^2} = 1 - \frac{n_e e^2}{2\epsilon_0 m_e \omega^2}. \tag{3.33}$$

The additional phase relative to vacuum through a plasma column is

$$\phi(\mathbf{x}) = \frac{2\pi}{\lambda} \int (n-1)\, dz \tag{3.34}$$

$$= -r_e \lambda \int n_e(\mathbf{x}, z)\, dz, \tag{3.35}$$

where $r_e = e^2/(4\pi\epsilon_0 m_e c^2)$ is the classical electron radius and $\mathbf{x}$ is the two-dimensional transverse coordinate. Thus, phase fluctuations satisfy

$$\boxed{\delta\phi(\mathbf{x}) = -r_e \lambda \int \delta n_e(\mathbf{x}, z)\, dz.} \tag{3.36}$$

In particular, the phase variance scales as λ^2 for a fixed density-fluctuation field.

3.4 Kolmogorov electron-density turbulence

Assume statistically homogeneous, isotropic fluctuations with a three-dimensional power spectrum

$$\boxed{P_{n_e}(q) = C_n^2 q^{-\beta}, \qquad \beta = \frac{11}{3},} \tag{3.37}$$

within the inertial range $q_{\rm out} \ll q \ll q_{\rm in}$. Here $q = |\mathbf{q}|$ is a spatial wavenumber; C_n^2 describes the turbulence strength. More generally, we retain $2 < \beta < 4$ until the final step.

Define the phase structure function between two transverse positions separated by $\boldsymbol{\rho}$:

$$D_\phi(\rho) = \left\langle [\phi(\mathbf{x}+\boldsymbol{\rho}) - \phi(\mathbf{x})]^2 \right\rangle . \tag{3.38}$$

The phase is a line-of-sight integral of δn_e. For a statistically homogeneous medium with propagation length L much greater than the turbulence correlation length, the longitudinal integration selects the $q_z \simeq 0$ part of the 3D spectrum. The resulting transverse phase power spectrum therefore scales as

$$P_\phi(\mathbf{q}_\perp) \propto (r_e\lambda)^2 L\, C_n^2 q_\perp^{-\beta}. \tag{3.39}$$

For a nonuniform path, replace $C_n^2 L$ with the scattering measure $\mathrm{SM} = \int C_n^2(z)\, dz$.

The structure function is related to this two-dimensional spectrum by

$$D_\phi(\rho) = 2\int d^2q_\perp\, P_\phi(q_\perp)\,[1 - \cos(\mathbf{q}_\perp \cdot \boldsymbol{\rho})] \tag{3.40}$$

$$\propto \lambda^2 \mathrm{SM} \int_0^\infty dq_\perp\, q_\perp^{1-\beta}\,[1 - J_0(q_\perp\rho)]\,. \tag{3.41}$$

Here J_0 is the Bessel function obtained by averaging the cosine over the polar angle. Substitute $u = q_\perp\rho$:

$$D_\phi(\rho) \propto \lambda^2 \mathrm{SM}\, \rho^{\beta-2} \underbrace{\int_0^\infty du\, u^{1-\beta}[1 - J_0(u)]}_{\text{constant for } 2<\beta<4}\,. \tag{3.42}$$

The integral converges: as $u \to 0$, $1 - J_0(u) \sim u^2/4$, and at large u the integral is convergent for $\beta > 2$. Therefore,

$$\boxed{D_\phi(\rho) \propto \lambda^2 \mathrm{SM}\, \rho^{\beta-2}.} \tag{3.43}$$

For Kolmogorov turbulence, $\beta = 11/3$, so

$$\boxed{D_\phi(\rho) \propto \lambda^2 \mathrm{SM}\, \rho^{5/3}.} \tag{3.44}$$

3.5 The diffractive scale

Define $r_{\rm diff}$ by a root-mean-square phase difference of one radian:

$$D_\phi(r_{\rm diff}) = 1. \tag{3.45}$$

Thus, at fixed scattering measure,

$$1 \propto \lambda^2 r_{\rm diff}^{\beta-2}, \qquad \boxed{r_{\rm diff} \propto \lambda^{-2/(\beta-2)}.} \tag{3.46}$$

For $\beta = 11/3$,

$$\boxed{r_{\rm diff} \propto \lambda^{-6/5} \propto \nu^{6/5}.} \tag{3.47}$$

The key difference from a fixed plasma lens is that lower frequencies sample *smaller* diffractive transverse scales. Consequently, the angular broadening grows faster than λ^2.

3.6 From diffractive scale to scattering angle

Two patches of a wavefront separated by $r_{\rm diff}$ have a phase difference of order one radian. Their characteristic transverse phase gradient is therefore $|\nabla_\perp \phi| \sim 1/r_{\rm diff}$. A wavevector is the phase gradient, and its longitudinal magnitude is approximately $k = 2\pi/\lambda$. Hence

$$\theta_{\rm scatt} \sim \frac{|\mathbf{k}_\perp|}{k} \sim \frac{\lambda}{2\pi r_{\rm diff}}. \tag{3.48}$$

Combining with the frequency dependence of $r_{\rm diff}$,

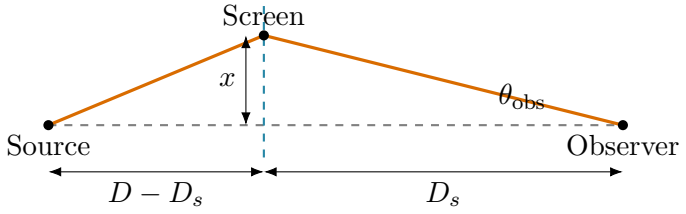
$$\boxed{\theta_{\rm scatt} \propto \lambda^{1+2/(\beta-2)} = \lambda^{\beta/(\beta-2)}.} \tag{3.49}$$

For Kolmogorov turbulence,

$$\boxed{\theta_{\rm scatt} \propto \lambda^{11/5} \propto \nu^{-11/5}.} \tag{3.50}$$

3.7 Geometric path difference and pulse broadening

Let the source–observer distance be D, with a screen at distance D_s from the observer. A scattered path passes through the screen at transverse displacement x.



The direct path has length D. The scattered path is

$$L_{\rm scatt} = \sqrt{(D - D_s)^2 + x^2} + \sqrt{D_s^2 + x^2}. \tag{3.51}$$

For $x \ll D_s, D - D_s$, use $\sqrt{a^2 + x^2} \simeq a + x^2/(2a)$ to obtain

$$\Delta L = L_{\rm scatt} - D \simeq \frac{x^2 D}{2D_s(D - D_s)}. \tag{3.52}$$

The observed angle is $\theta_{\rm obs} \simeq x/D_s$, so

$$\boxed{\Delta t = \frac{\Delta L}{c} \simeq \frac{D D_s}{2c(D - D_s)}\theta_{\rm obs}^2.} \tag{3.53}$$

Alternatively, the total ray deflection at the screen is $\theta_{\rm def} \simeq x/(D - D_s) + x/D_s$, yielding

$$\boxed{\Delta t \simeq \frac{D_s(D - D_s)}{2cD}\theta_{\rm def}^2.} \tag{3.54}$$

Both definitions give the same frequency dependence because the geometrical distances are frequency independent. In the standard strong-scattering thin-screen picture, the characteristic pulse-broadening time scales with this geometric delay:

$$\boxed{\tau_{\rm scatt} \propto \theta_{\rm scatt}^2.} \tag{3.55}$$

3.8 The Kolmogorov scattering index

Putting the pieces together,

$$P_{n_e}(q) \propto q^{-\beta}, \tag{3.56}$$
$$D_\phi(\rho) \propto \lambda^2 \rho^{\beta-2}, \tag{3.57}$$
$$r_{\rm diff} \propto \lambda^{-2/(\beta-2)}, \tag{3.58}$$
$$\theta_{\rm scatt} \propto \lambda^{\beta/(\beta-2)}, \tag{3.59}$$
$$\tau_{\rm scatt} \propto \lambda^{2\beta/(\beta-2)} \propto \nu^{-2\beta/(\beta-2)}. \tag{3.60}$$

Therefore the scattering index, defined by $\tau_{\rm scatt} \propto \nu^\alpha$, is

$$\boxed{\alpha = -\frac{2\beta}{\beta-2}.} \tag{3.61}$$

For a Kolmogorov spectrum, $\beta = 11/3$:

$$\boxed{\alpha = -\frac{2(11/3)}{11/3-2} = -\frac{22}{5} = -4.4.} \tag{3.62}$$

When does this prediction apply? The -4.4 result assumes isotropic Kolmogorov turbulence, a diffractive scale inside the inertial range, and strong scattering with geometric multipath delays dominating the broadening. Finite screen extent, anisotropy, inner-scale effects, multiple screens, or a different turbulence spectrum can change the observed frequency dependence. Cosmological redshift changes the observed time and frequency normalization for a fixed screen, but not this power-law exponent.

3.9 The Pulse Broadening Function

Radio waves propagating through an inhomogeneous plasma can follow multiple paths between a source and an observer. Because these paths have different geometric lengths, photons emitted simultaneously can arrive at different times. The resulting distribution of arrival-time delays is described by the **pulse broadening function** (PBF).

If the intrinsic pulse profile is $I_{\rm int}(t)$, the observed profile is approximately the convolution

$$\boxed{I_{\rm obs}(t) = \int_{-\infty}^{\infty} I_{\rm int}(t')\,{\rm PBF}(t-t')\,dt'.} \tag{3.63}$$

We will derive the PBF for an idealized thin scattering screen with an isotropic Gaussian distribution of scattering angles.

3.9.1 Geometric Time Delays

Consider a source at distance D from the observer, with a thin scattering screen located a distance D_s from the observer.

As derived previously, a ray arriving at an angle θ relative to the direct line of sight has an additional geometric path length

$$\Delta L \simeq \frac{D_s D}{2(D - D_s)}\theta^2. \tag{3.64}$$

The corresponding time delay is

$$t = \frac{\Delta L}{c}. \tag{3.65}$$

Defining the geometric factor

$$A \equiv \frac{D_s D}{2c(D - D_s)}, \tag{3.66}$$

we can write

$$\boxed{t = A\theta^2.} \tag{3.67}$$

This is the essential relationship: the geometric delay is proportional to the square of the angular offset.

Since $\theta^2 \geq 0$, geometric scattering delays are always non-negative relative to the direct path.

3.9.2 The Angular Distribution of Scattered Rays

Assume that the angular deflections in two transverse directions, θ_x and θ_y, are independent Gaussian random variables with zero mean and equal variance σ_θ^2:

$$p(\theta_x, \theta_y) = \frac{1}{2\pi\sigma_\theta^2} \exp\left[-\frac{\theta_x^2 + \theta_y^2}{2\sigma_\theta^2}\right]. \tag{3.68}$$

The total angular offset is

$$\theta^2 = \theta_x^2 + \theta_y^2. \tag{3.69}$$

Because the distribution is isotropic, it is convenient to transform into polar coordinates:

$$d\theta_x\, d\theta_y = \theta\, d\theta\, d\varphi. \tag{3.70}$$

Integrating over the azimuthal angle φ,

$$p_\theta(\theta)\, d\theta = \int_0^{2\pi} p(\theta_x, \theta_y)\, \theta\, d\varphi\, d\theta \tag{3.71}$$

$$= \frac{\theta}{\sigma_\theta^2} \exp\left(-\frac{\theta^2}{2\sigma_\theta^2}\right) d\theta. \tag{3.72}$$

Thus the probability density for the magnitude of the angular offset is

$$\boxed{p_\theta(\theta) = \frac{\theta}{\sigma_\theta^2} \exp\left(-\frac{\theta^2}{2\sigma_\theta^2}\right), \qquad \theta \geq 0.} \tag{3.73}$$

This is a Rayleigh distribution.

3.9.3 Converting Angular Deflections into Time Delays

We now transform the angular probability distribution into a distribution of arrival times.

Recall that

$$t = A\theta^2. \tag{3.74}$$

Solving for θ,

$$\theta = \sqrt{\frac{t}{A}}. \tag{3.75}$$

Differentiating,

$$\frac{d\theta}{dt} = \frac{1}{2A\theta}. \tag{3.76}$$

Conservation of probability requires

$$p_t(t)\,dt = p_\theta(\theta)\,d\theta. \tag{3.77}$$

Therefore,

$$p_t(t) = p_\theta(\theta)\left|\frac{d\theta}{dt}\right|. \tag{3.78}$$

Substituting the angular distribution,

$$p_t(t) = \frac{\theta}{\sigma_\theta^2}\exp\left(-\frac{\theta^2}{2\sigma_\theta^2}\right)\frac{1}{2A\theta} \tag{3.79}$$

$$= \frac{1}{2A\sigma_\theta^2}\exp\left(-\frac{t}{2A\sigma_\theta^2}\right). \tag{3.80}$$

Define the characteristic scattering timescale

$$\boxed{\tau_{\text{scatt}} \equiv 2A\sigma_\theta^2.} \tag{3.81}$$

Using our expression for A,

$$\boxed{\tau_{\text{scatt}} = \frac{D_s D}{c(D - D_s)}\sigma_\theta^2.} \tag{3.82}$$

The probability distribution of arrival times becomes

$$\boxed{p_t(t) = \frac{1}{\tau_{\text{scatt}}}\exp\left(-\frac{t}{\tau_{\text{scatt}}}\right), \qquad t \geq 0.} \tag{3.83}$$

For negative delays, the probability is zero. We express this using the Heaviside step function,

$$H(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0. \end{cases} \tag{3.84}$$

Our final pulse broadening function is therefore

$$\boxed{\mathrm{PBF}(t) = \frac{1}{\tau_{\mathrm{scatt}}} e^{-t/\tau_{\mathrm{scatt}}} H(t).} \tag{3.85}$$

The normalization can be verified directly:

$$\int_{-\infty}^{\infty} \mathrm{PBF}(t)\, dt = \int_{0}^{\infty} \frac{1}{\tau_{\mathrm{scatt}}} e^{-t/\tau_{\mathrm{scatt}}} dt = 1. \tag{3.86}$$

3.9.4 Why Scattered Pulses Have Exponential Tails

Suppose the source emits an infinitely narrow pulse,

$$I_{\mathrm{int}}(t) = F\, \delta(t), \tag{3.87}$$

where F is the pulse fluence.

Convolving this with the PBF gives

$$I_{\mathrm{obs}}(t) = F \int_{-\infty}^{\infty} \delta(t')\, \mathrm{PBF}(t - t')\, dt' \tag{3.88}$$

$$= F\, \mathrm{PBF}(t). \tag{3.89}$$

Thus,

$$\boxed{I_{\mathrm{obs}}(t) = \frac{F}{\tau_{\mathrm{scatt}}} e^{-t/\tau_{\mathrm{scatt}}} H(t).} \tag{3.90}$$

An intrinsically instantaneous pulse is observed as a one-sided exponential tail.

For a pulse with finite intrinsic width, the observed profile is the convolution of the intrinsic emission and this exponential response.

For example, if the intrinsic pulse is Gaussian,

$$I_{\mathrm{int}}(t) = \frac{F}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(t - t_0)^2}{2\sigma^2}\right], \tag{3.91}$$

then convolution with the exponential PBF gives

$$\boxed{\begin{aligned} I_{\mathrm{obs}}(t) = \frac{F}{2\tau_{\mathrm{scatt}}} \exp\left[\frac{\sigma^2}{2\tau_{\mathrm{scatt}}^2} - \frac{t - t_0}{\tau_{\mathrm{scatt}}}\right] \\ \times \mathrm{erfc}\left[\frac{1}{\sqrt{2}}\left(\frac{\sigma}{\tau_{\mathrm{scatt}}} - \frac{t - t_0}{\sigma}\right)\right]. \end{aligned}} \tag{3.92}$$

Here erfc is the complementary error function.

This is the familiar exponentially modified Gaussian pulse profile, commonly used to model temporally scattered radio bursts.

3.9.5 Frequency Dependence

The characteristic scattering timescale depends on the angular distribution of the scattered radiation.

Since

$$\tau_{\rm scatt} \propto \sigma_\theta^2, \tag{3.93}$$

the frequency dependence of the PBF timescale follows from the frequency dependence of the scattering angle.

For an ideal Kolmogorov turbulent plasma in the strong-scattering regime,

$$\sigma_\theta \propto \nu^{-11/5}, \tag{3.94}$$

and therefore

$$\boxed{\tau_{\rm scatt} \propto \nu^{-22/5} = \nu^{-4.4}.} \tag{3.95}$$

In practice, scattering timescales are often parameterized relative to a reference frequency ν_0:

$$\boxed{\tau_{\rm scatt}(\nu) = \tau_0 \left(\frac{\nu}{\nu_0}\right)^{\alpha}.} \tag{3.96}$$

The scattering index α can then be measured from observations across multiple frequency bands.

3.9.6 Limitations of the Exponential PBF

The one-sided exponential PBF derived here relies on an isotropic Gaussian distribution of scattering angles and an idealized thin-screen geometry.

A real turbulent plasma does not necessarily produce an exactly Gaussian angular distribution. In particular, an ideal Kolmogorov phase structure function is not quadratic, so its angular scattering distribution and corresponding PBF are generally not exactly exponential.

Anisotropic scattering, multiple scattering screens, finite screen sizes, and distributed scattering media can also produce different pulse broadening functions.

Consequently, an exponential PBF is a useful model, but it is not a universal description of radio-wave scattering.